



FAILSAFE



Superscript



base

THE 2026 AI X BLOCKCHAIN CONVERGENCE REPORT

HOW ASIA IS SHAPING THE
CONVERGENCE FRONTIER

Grab

Virtual
PROTOCOL

coinbase

 KAITO

aws

Sogni

chatandbuild

 Aethir

Foreword	1
Contributor Notes	1
1. Executive Summary	3
2. The Current Snapshot of Crypto x AI	5
3. Why Asia? The Macro Context	7
4. The Global Landscape: AI & Blockchain Convergence	10
5. Examples of AI & Blockchain Convergence in Asia	14
6. The Road to 2026 and Beyond	23
7. Risk, Governance, and Responsible Acceleration	28
8. Conclusion	31
Appendix & Partner Notes	32
Methodology	34

Foreword

By Erik Reppel

Head of Engineering, Coinbase Developer Platform and creator of x402

The Asia-Pacific ecosystem is emerging as one of the most active testing grounds for the convergence of AI and Blockchain. Developers, regulators, and platforms across the region are experimenting with autonomous agents in finance, logistics, gaming, and identity tied to blockchain infrastructure, while balancing speed with responsible deployment.

AI agents are moving from experimental interfaces to fully capable systems that can perceive, reason, and act. To operate safely in real economic environments, these agents need reliable ways to move value, verify identity, and anchor their decisions in transparent systems. This is where the intersection of AI and blockchain becomes technically meaningful rather than aspirational.

x402 was designed to address a simple engineering constraint: agents need a standardized, interoperable way to request and settle payments across networks. By treating value transfer as a protocol rather than an application feature, x402 provides a foundation for agents to interact with wallets, chains, and payment services without brittle integrations. In parallel, blockchains supply verifiable state and auditability, while stablecoins and L2 networks provide predictable settlement. This combination creates the conditions for agentic systems to operate safely at scale.

This report highlights how APAC teams are implementing real-world systems that combine AI reasoning with on-chain execution, often under complex regulatory and infrastructural constraints.

Contributor Notes

Jesse Pollack

CEO & Creator of Base

“The powerful thing about AI and blockchain is that they’re both impactful trends emerging at the same time and are very additive to each other ... One challenge with AI is proving authenticity — is this a human or a bot? Agents also need a way to transact with money, which is difficult with legacy systems. Crypto solves these problems. Blockchain is a powerful tool for proving authenticity ... I expect these trends will drive innovation, as we’ll extend human capacity with agents that now also have superpowers like handling money and proving authenticity.” ([Source](#))

Philipp Kandal

Chief Product Officer, Grab

“We’re excited about the developments in these spaces. AI has been part of Grab’s DNA from the very beginning, and we are always exploring how to harness the latest technologies to solve uniquely Southeast Asian challenges. We intend to continue playing a key role supporting Southeast Asia’s innovation journey.”

Looking forward

Together, these perspectives point to a clear trajectory: AI will increasingly act, blockchains will increasingly verify, and APAC will remain a key region proving how these systems can work safely and effectively in production.

1. Executive Summary

The year 2025 marked a global tipping point for the convergence of AI and blockchain technologies, but no region embodies its potential more than Asia Pacific. Here, a young, mobile-native population, progressive regulators, and deep capital pools are fueling experiments that will define digital commerce through 2026 and beyond.

AI brings pattern recognition and autonomy; blockchain adds provenance, verifiability and ownership. Together, they enable trustless intelligence flows: agents that can meaningfully transact, markets that self-govern, data that is verified and access to models for all.

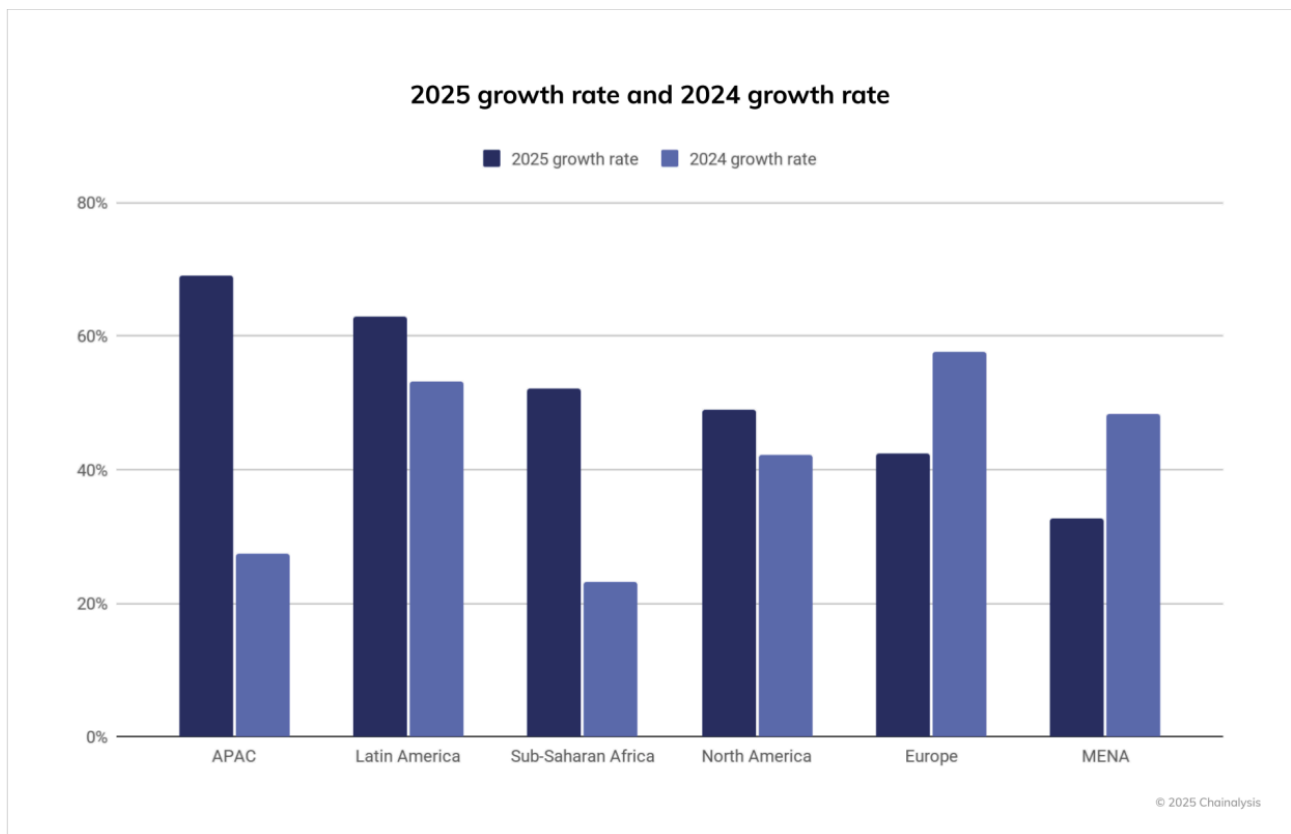


Figure 1: YoY On-Chain Activity Growth by Region

(Source: [2025 Geography of Crypto Report, Chainalysis](#))

According to [Messari's Sizing the Crypto x AI Market \(2025\)](#), the sector already exceeds \$2.7 billion in tokenized value, with Asia accounting for nearly 40 % of active projects. [Chainalysis](#) data shows Asia-Pacific was the fastest-growing region for on-chain activity in 2025 (+69 % YoY), while AI venture funding in Asia hit record highs (57.9% of global VC allocations in Q1 2025 per [PitchBook](#)).

The space where AI meets crypto is a nascent one and perhaps one which has already gone through one wave of speculation. That said, there is real potential for both tech trends to augment one another and it is most likely to create value when it relies on the things that blockchains do best ([Superscript investment thesis via Jacob Ko interview](#)):

- Incentivizing data collection & augmentation with incentives & tokens
- Cross-border payment rails for autonomous agents
- Coordination of scarce compute resources for AI workloads
- Provenance and tracking of data and models used in inference

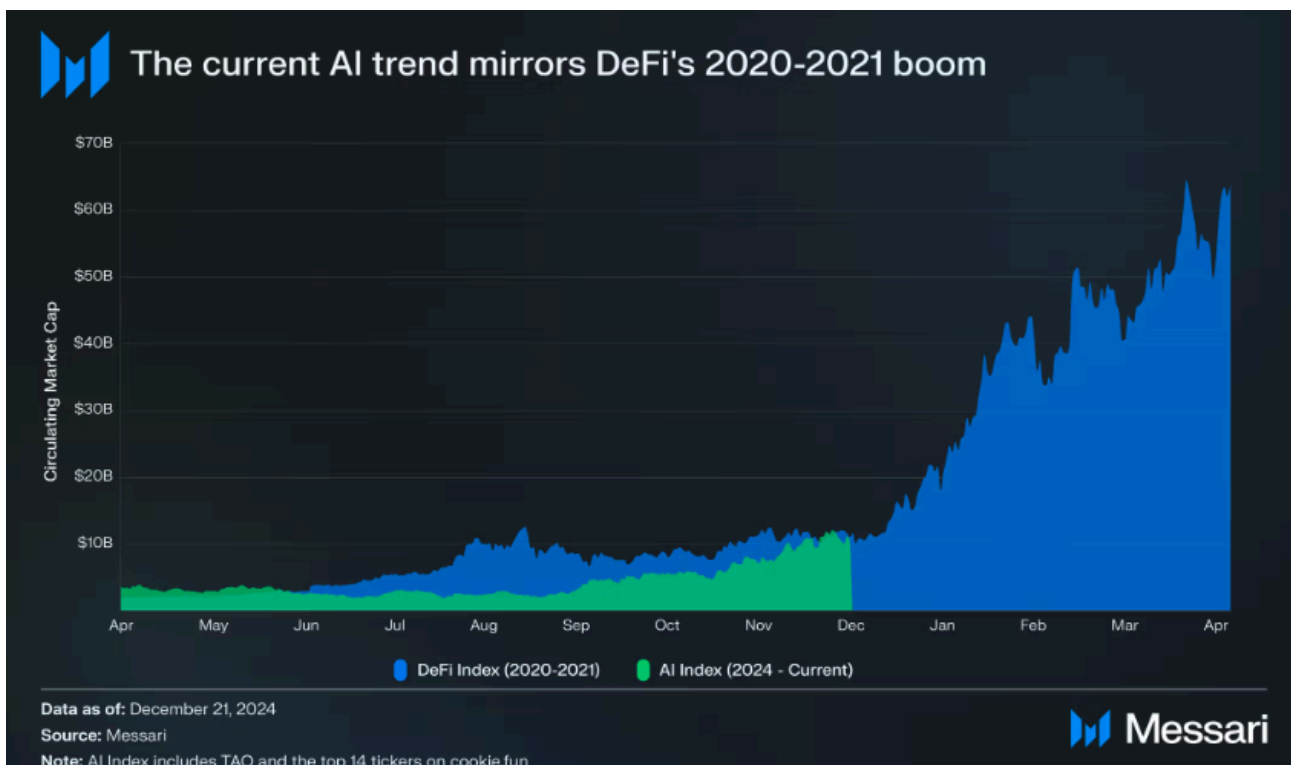


Figure 2: AI Trend (Dec 24) mapped against DeFi Trend of (2020-21)

(Source: [Messari](#))

The report unpacks why Asia is becoming the convergence epicenter, examines key use cases, profiles flagship projects, and addresses the emerging risk and governance imperatives that will shape the road to 2026.

2. The Current Snapshot of Crypto x AI

Crypto and AI intersect across a wide range of layers in the modern AI /LLM stack. From how data is collected to how models are trained and coordinated, Web3 protocols are being applied to many of the core bottlenecks in AI infrastructure and deployment. These projects span the full AI value chain, each tackling a different coordination, trust, or incentive problem that emerges as AI systems take over our world.

We attempted to layout the current landscape of some of the top Crypto x AI projects currently operating across these areas:

The AI LLM Value Chain & Crypto x AI Protocols			
		Description	Crypto x AI Protocols
Data	Collection	Protocols that generate or gather raw data from users, devices, or the physical world	 PrismaX  Hivemapper  Grass  375ai  Poseidon  Chakra
	Storage	Decentralized systems for storing large data sets reliably over time	 Filecoin  Arweave  Irys  Walrus  Zero Gravity
	Labelling	Human or agent-driven annotation turning raw data into training-ready data	 Sapien  Sahara  Perle  Codddata
	Financialization	Protocols that turn data into an economic asset with ownership, access control & incentives	 Vana  Ocean Protocol
Training	Foundation / Gradient	Direct training where model weights are updated using gradients on datasets	 Gensyn  Nous  Prime Intellect  Flock.io
	Feedback / Emergent	Model improvement driven by rewards, evaluations, or collective feedback loops	 Prime Intellect  Ambient  Fortytwo
	Orchestration	Coordination layers that manage contributors, processes & ownership around training	 Plurals  Templar (Bittensor Subnet)
Compute	Cloud	Dedicated, schedulable GPU infrastructure provisioned specifically for AI workloads	 Akash  Render  io.net  Hyperbolic  EigenCloud  Aethir  Exabits
	Edge	Compute sourced from non-dedicated, device-native hardware like phones, laptops	 exo  Pin AI
	Financialization	Protocols that underwrite, tokenize, or finance compute capacity	 GAIB  USD.AI
Inference	Inference as a Service	APIs that return model outputs on demand to developers or applications	 Prodia  Gradient  Chutes  Distribute AI
	Trust-minimized	Inference systems designed to reduce reliance on trusting a single operator	 Ambient  Fortytwo
Model Coordination	Market-based	Competitive systems where models are ranked and rewarded based on performance	 Bittensor  Numerai  Post Fiat  TIG
	Access Coordination	Protocols that govern which models can be used, by whom, and under what conditions	 Ritual  EigenCloud  Sentient  Near
	Signal Aggregation	Systems that combine outputs from many models into a collective signal	 Allora  FractionAI  Fortytwo

AI agents is another area where AI and blockchains overlap with blockchains powering payments and differing solutions in identity and verification. As AI Agents become more autonomous, in their ability to execute tasks, manage capital, and interact with other agents, blockchains provide the key primitives these systems need.

These projects sit at the intersection of execution, payments, and coordination, making it a key part of the Crypto x AI landscape today:

AI Agent framework & Crypto x AI projects		
	Description	Crypto x AI Projects
Frameworks	SDKs and tools for defining agent logic, planning and use	 ElizaOS  Dev.fun
Runtime	Execution environments where agents run persistently with keys, state, and scheduling	 KiteAI  Talus  Near
ID & Permission	Systems that define who an agent is and what it's allowed to do	 World ID  Humanity Protocol
Agent Coordination	Protocols that govern how agents interact, delegate, or cooperate	 Theoriq  Gala  KiteAI  Talus  Virtuals
Task Routing	Marketplaces and routers that assign work to agents	 Virtuals  OpenSerV  Symphony  Creator.bid
Payments / Wallet Infra	Rails that let agents pay, get paid, and manage budgets autonomously	 x402  Radius  Cookie DAO  Payman
Analytics	Monitoring and observability tools for agent performance and behavior	 Kaito.ai  TrueNorth
Agent Security	Guardrails, monitoring, and protection against exploits or misuse	 Sherlock  Octane  Forta
Applications	End-user products where agents perform real tasks for people	 Giza  Chatand Build  Butler (Virtuals)  Donut

3. Why Asia? The Macro Context

Digital Demographics and Adoption Velocity

Asia hosts over half of the world's internet users and 1.7 billion smartphone owners. Markets like India, Indonesia, Vietnam, and the Philippines leapfrogged desktop computing straight into super-app ecosystems (Grab, WeChat, Gojek). This mobile-first culture creates an ideal sandbox for AI and blockchain applications: from AI-driven micro-investing wallets to tokenized game assets.

Crypto & AI Nexus

Asia's rise in grassroots crypto use is clear: India, Vietnam, Indonesia, and the Philippines all rank in the top 10 of Chainalysis' 2025 Crypto Adoption Index, ahead of long-established Western economies like the United Kingdom and a number of European markets. In India specifically, Reuters ([Feb 2025](#)) reports that young people are increasingly turning to crypto trading to supplement stagnant wages and limited job opportunities. This socioeconomic pressure contributes to faster and broader grassroots adoption across the region.

At the same time, Asia produces the world's largest pool of AI engineers and patents, with China alone accounting for 69.7% of all granted AI patents ([2025 AI Index Report](#)). Chinese AI developers are also achieving global traction through highly cost-efficient model architectures that rely on smaller, compute-optimized models and older hardware ([Nov 2025, AI Jazeera](#)). This efficiency dramatically lowers the cost of deploying AI at scale, enabling faster and wider adoption across Asian markets, especially among startups and cost-constrained enterprises.

Policy and Regulatory Catalysts

Singapore's Monetary Authority (MAS) has pioneered projects like [Project Guardian](#) to enhance liquidity and efficiency of financial markets through asset tokenisation and [Veritas](#) to strengthen internal governance around the application of AI and the management and use of data. In Japan, the focus is on long-term structural growth: On the crypto front, the government is advancing [regulatory reforms](#) to reclassify crypto as investment instruments and planning a [major overhaul of the tax regime](#), potentially lowering rates to 20% to boost competitiveness. At the same time, the government has released an [AI white paper](#) outlining an ambitious roadmap to make Japan the world's most "AI-friendly" country, signalling strong commitment to promoting AI research, infrastructure and regulation. In Hong Kong, the [2024 licensing and regulatory reforms](#) by the SFC, which ended transitional arrangements and required all virtual-asset trading platforms to be fully licensed, [revitalised the city's Web3 ecosystem](#) and reopened it to regulated blockchain-enabled innovation.

Venture and Corporate Momentum

Several regional investors have publicly backed AI-blockchain ventures. For example, early-stage AI-powered Web3 search-engine [Kaito raised US\\$5.5 million in a Series A](#) led by Superscript and Spartan. Meanwhile, HashKey Group [recently launched a dedicated AI unit](#), HashKey AI, rolling out its first Web3-AI product ModAI for blockchain-native AI applications. [PitchBook's Q1 2025 Venture Monitor](#) shows that AI and ML companies accounted for 71.1% of VC deal value in the \$50M+ deal-size bucket, highlighting how large-scale capital is increasingly concentrating around AI. This tilt toward bigger AI rounds reflects a strong shift in investor priority and underscores why regional funds are accelerating exposure to AI-linked blockchain projects.

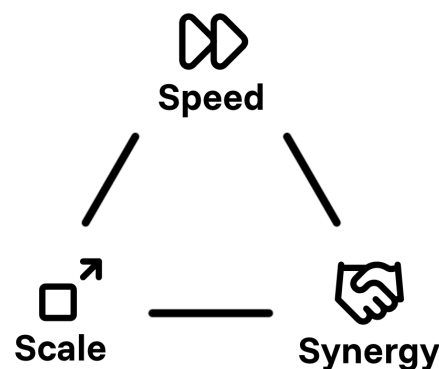


Figure 3: Asia's Trifecta

Asia's Trifecta: Scale (huge user base and data pool) + Speed (cultural agility for tech adoption) + Synergy (public-private collaboration).

In summary, Asia's macro context is characterized by scale, speed, and synergy:

- **Scale:** Unparalleled user numbers and data generation (fuel for AI models) combined with large transaction volumes on blockchains. Asia-Pacific is home to 4.3 billion people, representing 60% of the world's population ([UNFPA, 2025](#)). With 66% internet penetration translating to approximately 2.8 billion internet users ([ITU, 2025](#)). This massive digital population generates enormous data volumes that feed AI model training, while the region's crypto adoption rates create substantial on-chain transaction activity, providing rich datasets for AI and blockchain convergence applications.

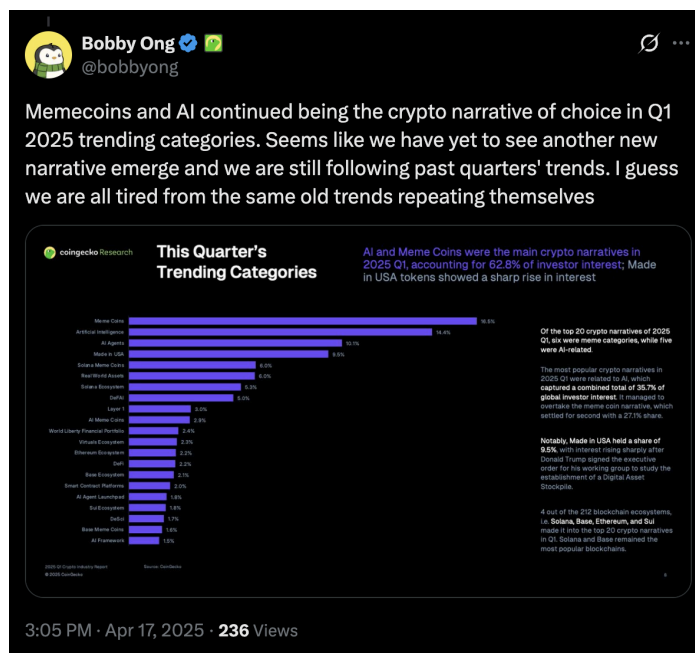


Figure 4: Bobby Ong's X Tweet Aug 2025

(Source: [X](#))

- Speed:** Markets in Asia tend to be open to investing in new tech trends. When AI-themed crypto tokens became popular, exchanges in Asia saw trading volumes spike and local investors were among the first movers. *"Memecoins and AI continued being the crypto narrative of choice in Q1 2025...we have yet to see another new narrative emerge,"* noted CoinGecko's COO Bobby Ong in early 2025 (via X, reporting on [CoinGecko's Q1 Industry Report](#)). While some of this is speculative in nature, Asia's crypto community tends to respond quickly and we expect this trend to continue in the region as more crypto AI projects launch with better features and utility that start gaining product market fit.
- Synergy:** Collaboration between governments, corporates, and startups in Asia creates a supportive innovation ecosystem underpinned by strong institutional readiness. Several Asian governments rank highly on the Oxford Insights Government AI Readiness Index, particularly in governance, tech-sector, and data/infrastructure pillars, demonstrating the region's robust capacity to support ambitious AI and blockchain experimentation ([Oxford Insights, 2024](#)). This readiness translates into active public-private partnerships, such as the Global Finance & Technology Network (GFTN), a not-for-profit launched by MAS in 2024 that convenes regulators, banks, and tech firms across Asia to build "financial corridors" and pilot emerging technologies. Such platforms mean that when a new AI and blockchain solution is tested, such as decentralized identity for KYC or AI for fraud detection in crypto exchanges, there is often institutional support to validate and scale it. As the GFTN manifesto puts it: "Regulators inform innovators, and innovators inspire regulators... By staying above the fray, we create an environment where authenticity thrives, collaboration is frictionless, and confidence is built through transparency and shared purpose" ([GFTN MayManifesto](#)). This combination of institutional readiness and principled partnership in Asia is a key reason convergence technologies can advance rapidly in the region.

4. The Global Landscape: AI & Blockchain Convergence

While Asia leads the charge, the convergence of **artificial intelligence (AI)** and **blockchain** is a global phenomenon reshaping industries from finance to entertainment. The following overview situates Asia's leadership within this broader international context.

Market Scale and Composition

AI Crypto Market Cap Distribution - 2025-11-01

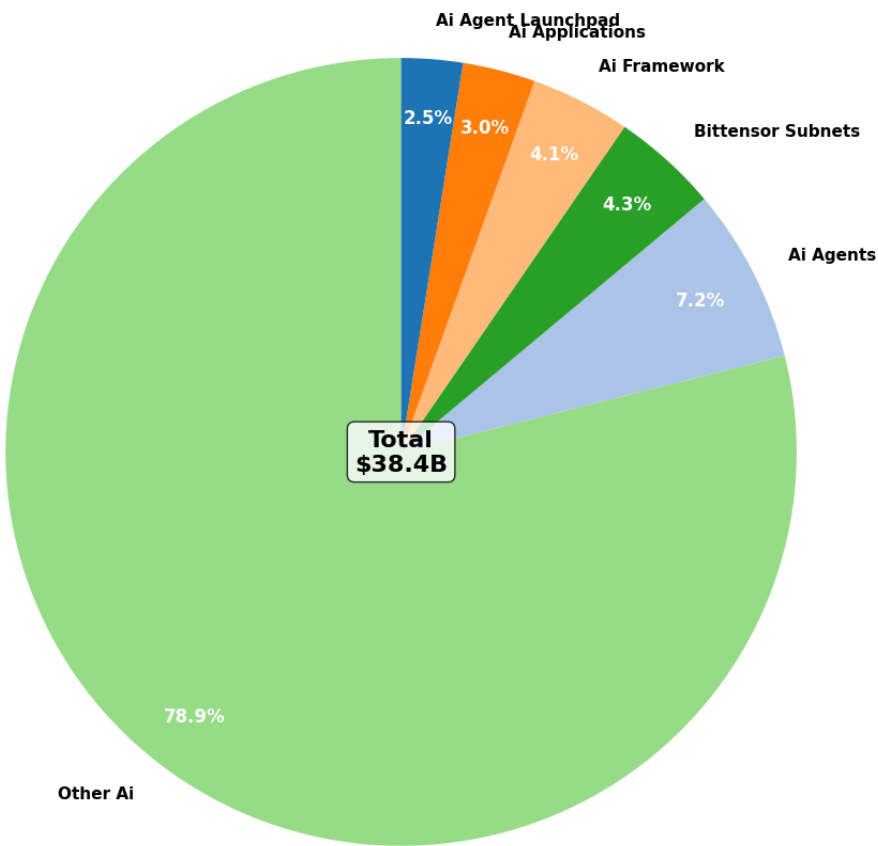


Figure 5: AI Crypto Market Cap Distribution

(Source: [Coin Gecko](#))

The AI & Blockchain sector has reached \$38.4 billion in market capitalization as of November 2025, though its composition reveals an industry still in early formation (Figure 1). Nearly 79% of this value sits in a broad "other-ai" category for tokens that relate to AI but don't exhibit specific convergence primitives. The specialized subcategories paint a more revealing picture: AI agents command \$2.76 billion (7.2%) as the largest defined segment, followed by Bittensor subnets at \$1.68 billion (4.3%), AI frameworks at \$1.57 billion (4.1%), AI applications at \$1.15 billion (3.0%), and AI agent launchpads at \$0.97 billion (2.5%).

This distribution, with substantial total capital yet fragmented deployment, reflects a market in transition. The dominance of undifferentiated AI tokens suggests speculative positioning, while the emergence of distinct infrastructure categories (agents, frameworks, launchpads) signals the beginning of product-market fit discovery. The actual convergence infrastructure stack totals only \$5.3 billion combined, indicating significant room for growth as genuine utility cases mature. Markets have signaled interest, but the infrastructure to deliver on that promise is only now being built.

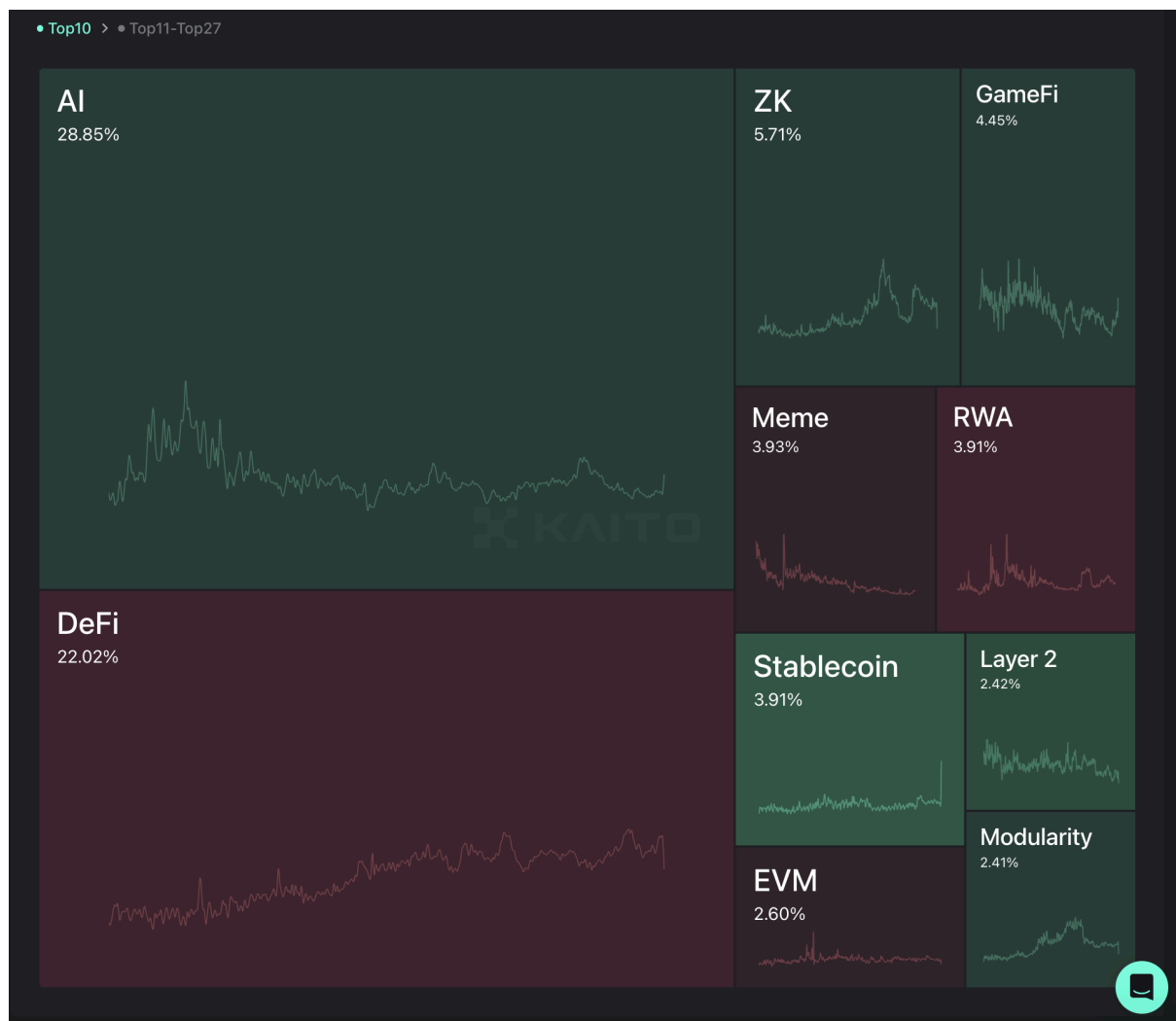


Figure 6: Crypto x AI Narrative Mindshare

(Source: [Kaito.ai](https://kaito.ai))

According to Kaito AI's narrative mindshare data, Artificial Intelligence (AI) has been the #1 narrative in mindshare for the past year - consistently outperforming DeFi, Memes, ZK, and every other sector. The momentum is being pulled forward by the broader AI wave: the pace of model improvements, agentic developments, mainstream consumer AI products, and a surge in global AI CapEx.

Global Investment and Startup Activity

Between 2024 and 2025, global investment in AI-blockchain ventures surged. Early pioneers such as [Fetch.ai](#) (UK) and [SingularityNET](#) (global) laid the groundwork for decentralized AI networks.

New entrants, including [OpenGradient](#) (US), which raised \$8.5M in 2024 for decentralized AI compute, reflect a wave of specialized infrastructure projects.

Major venture firms such as [a16z](#), [Pantera Capital](#), and [Coinbase Ventures](#) now back convergence startups, often alongside Asian funds, underscoring a shared global thesis.

Tech Giants and Institutional Entrants

Established technology leaders are weaving blockchain into their AI strategies:

- **IBM** uses blockchain to secure AI-driven supply chain insights, ensuring provenance and accountability.
- Warner Music Group partners with **Polygon Labs** to accelerate [decentralized music projects](#).

In Europe, the [European Blockchain Services Infrastructure \(EBSI\)](#) explores blockchain-backed data sharing, while the [EU AI Act](#) and [MiCA](#) together set a governance model emphasizing transparency and interoperability.

Global Use Case Trends

- **Decentralized Identity & Data Ownership:** Blockchain-based identity standards such as [Hyperledger Indy](#) and [W3C DIDs](#) are being merged with AI verification systems, enabling privacy-preserving data sharing. In Asia, initiatives like [Affinidi](#) (Singapore) and [Metadium](#) (Korea) are leading this charge.
- **AI Marketplaces & Data Exchange:** Platforms like [Ocean Protocol](#) and [Fetch.ai](#) enable tokenized data sharing and AI model monetization. These systems let data owners contribute to AI training securely, breaking big tech's data monopoly.
- **Autonomous Agents & AI DAOs:** The rise of “agentic” systems, where AI programs transact and collaborate autonomously, marks a pivotal shift. Experiments like [dTrade's](#) algorithmic traders, DAO-managed funds, and [Coinbase's AgentKit](#) (now evolved into x402) are early indicators of AI becoming an active economic participant.
- **AI-Enhanced Finance & DeFi:** Financial institutions integrate AI for predictive analytics and blockchain for transparent settlement. [Project Guardian](#), a Singapore-led collaboration, saw banks using AI to detect on-chain anomalies, improving DeFi compliance and fraud prevention.

Competitive and Collaborative Dynamics

Asia's rise coincides with an increasingly intertwined global ecosystem. Silicon Valley firms look east for innovation velocity and user adoption. Partnerships such as [Sony's \\$3.5M investment in Startale Labs](#) (Japan) reflect cross-border alignment in blockchain and entertainment IP. Conversely, Western AI firms collaborate with Asian exchanges like [Binance](#) and [OKX](#) to deploy AI-enhanced compliance and trading systems.

Yet fragmentation persists. China leads globally in AI development, producing 69.7% of all granted AI patents and open-source models like DeepSeek that have achieved international traction, but its [ban on cryptocurrency](#) limits how these capabilities can integrate with open, permissionless blockchain infrastructure. Asia's diversity thus mirrors a global 'splinternet' where AI leadership and blockchain adoption don't always overlap.

Global Regulation and Innovation Hubs

Jurisdictions such as [Dubai](#), [Singapore](#), and [Switzerland](#) serve as regulatory sandboxes where convergence experiments accelerate under structured oversight. Europe's rigorous privacy regime may slow rollout, but it provides a governance model others adapt regionally.

The global consensus is clear: AI and blockchain together solve trust and verification challenges that neither can address alone. From decentralized data markets in Europe to autonomous finance pilots in Singapore, real-world validation is accelerating.

5. Examples of AI & Blockchain Convergence in Asia

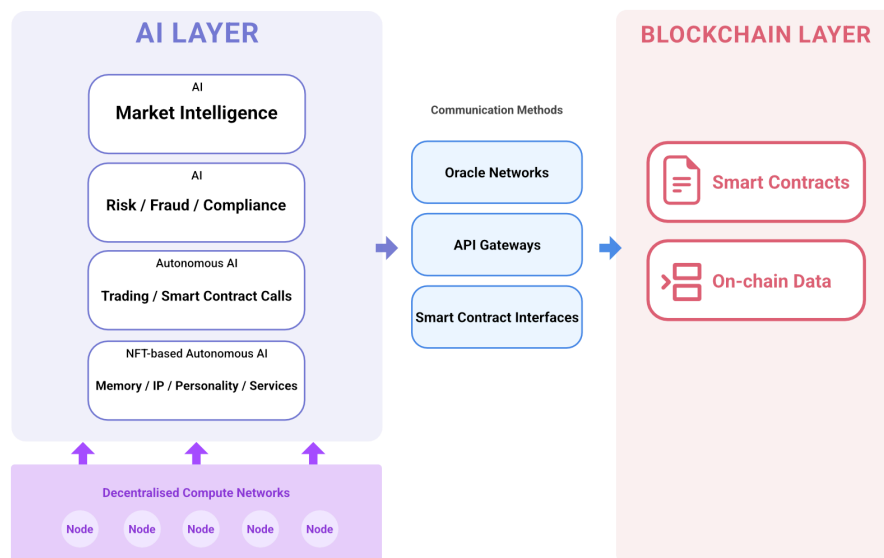


Figure 6: How AI converges with Blockchains

In this section, we explore specific case studies that illustrate how blockchain is being used in tandem with AI with a focus on projects tied to Asia or impacting the Asian landscape. These examples highlight practical applications of blockchain properties: from imbuing autonomous agents with identifiers & the ability to execute payments to incentivizing the provision of idle GPU compute resources via tokens and lastly using blockchain's provenance properties to solve identity & IP ownership issues.

Autonomous AI Agents

[Virtuals.io](https://www.virtuals.io) is an emerging protocol designed for deploying and scaling autonomous AI agents on-chain. Built originally on [Coinbase's Base network](https://www.coinbase.com/learn/what-is-crypto/a-guide-to-crypto), it has evolved into a full-stack framework for what the team calls *Agentic GDP*, defined as the economic output generated when humans, AI agents, and machines collaborate through blockchain systems.

The protocol is organized around four interoperable components:

- **Agent Commerce Protocol (ACP):** a standardized layer for agent-to-agent transactions and value exchange
- **Butler (social interface):** a conversational interface that connects humans and agents across networks like X and Telegram
- **Unicorn (liquidity layer):** a funding system for new agent-based applications and tokenized economies
- **Virtuals Robotics:** an integration layer that extends agents into real-world applications and devices

Virtuals enables developers to create AI agents with on-chain identities, wallets, and executable logic, allowing them to transact, earn, and collaborate autonomously. Examples include design agents that can generate media, process payments, and hire other agents for subtasks without human oversight.

This concept has strong relevance in Asia, where high digital literacy and mobile-native communities enable grassroots entrepreneurship. Virtuals envisions communities in India, Indonesia, and the Philippines co-owning and operating AI-driven “micro-enterprises” such as trading bots, media generators, or e-commerce assistants that run continuously on-chain. Its native token, [SVIRTUAL](#), reflects growing developer and investor interest, regularly seeing multimillion-dollar daily trading volume.

Agentic Commerce

In May 2025, Coinbase unveiled [x402](#), an evolution of its earlier AgentKit, marking a leap forward in what's now called agentic commerce, where AI systems act as autonomous financial participants, leveraging on-chain financial technology. While AgentKit gave AI agents a crypto wallet and basic transaction capability, x402 has evolved to be a secure, programmable HTTP-driven payments protocol for intelligent agents to transact, accessing external web resources autonomously.

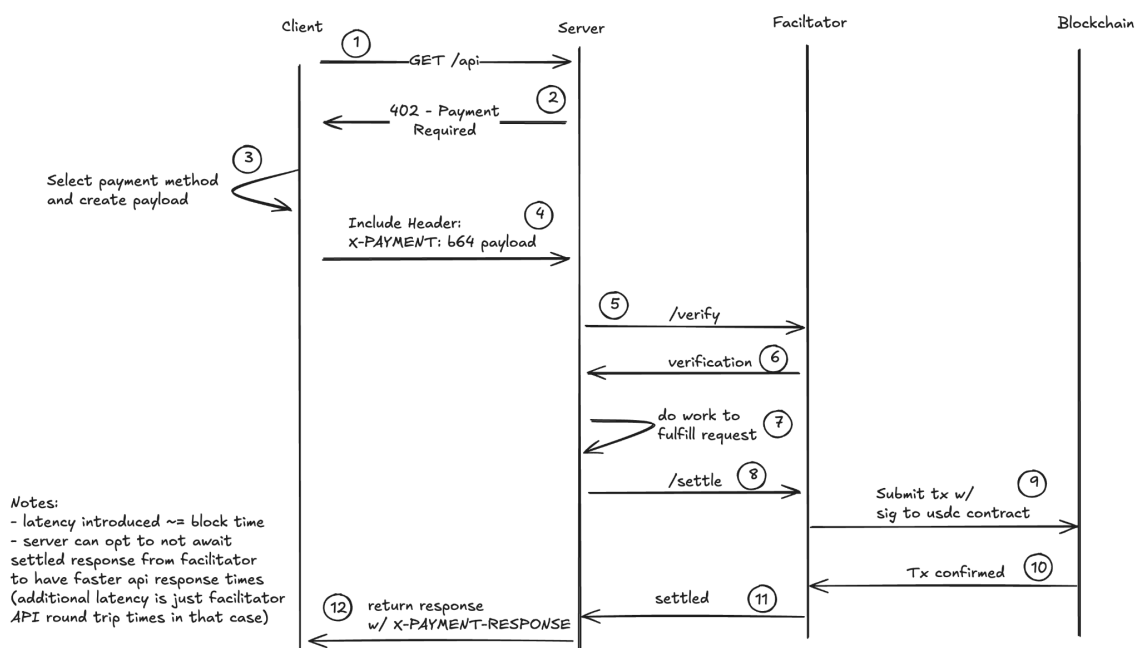


Figure 7: Flow of a payment using the x402 protocol

(Source: [x402 Github](#))

x402 is an open protocol that enables AI agents or applications to make automated, on-demand payments for digital services using stablecoins like USDC. It provides a lightweight, developer-friendly way for agents to pay for APIs, data, or other resources without relying on traditional payment rails, while maintaining low friction, micropayment support, and transparency. Combined with other Coinbase developer tools, x402 allows builders to create agent-driven workflows where software can transact, monetize services, or collaborate with other agents seamlessly.

Across Asia, x402 is already influencing early agentic deployments. An APAC startup, [Questflow](#) uses the x402 protocol to enable autonomous, pay-per-use payments between AI agents, where each micro-interaction in a swarm workflow triggers a USDC settlement on-chain without subscriptions, API keys, or manual human configuration. These implementations demonstrate how AI autonomy plus blockchain verifiability can enable a new economic paradigm: one where algorithms aren't just tools but economic actors, actively transacting, governing, and earning on behalf of their human or organizational creators.

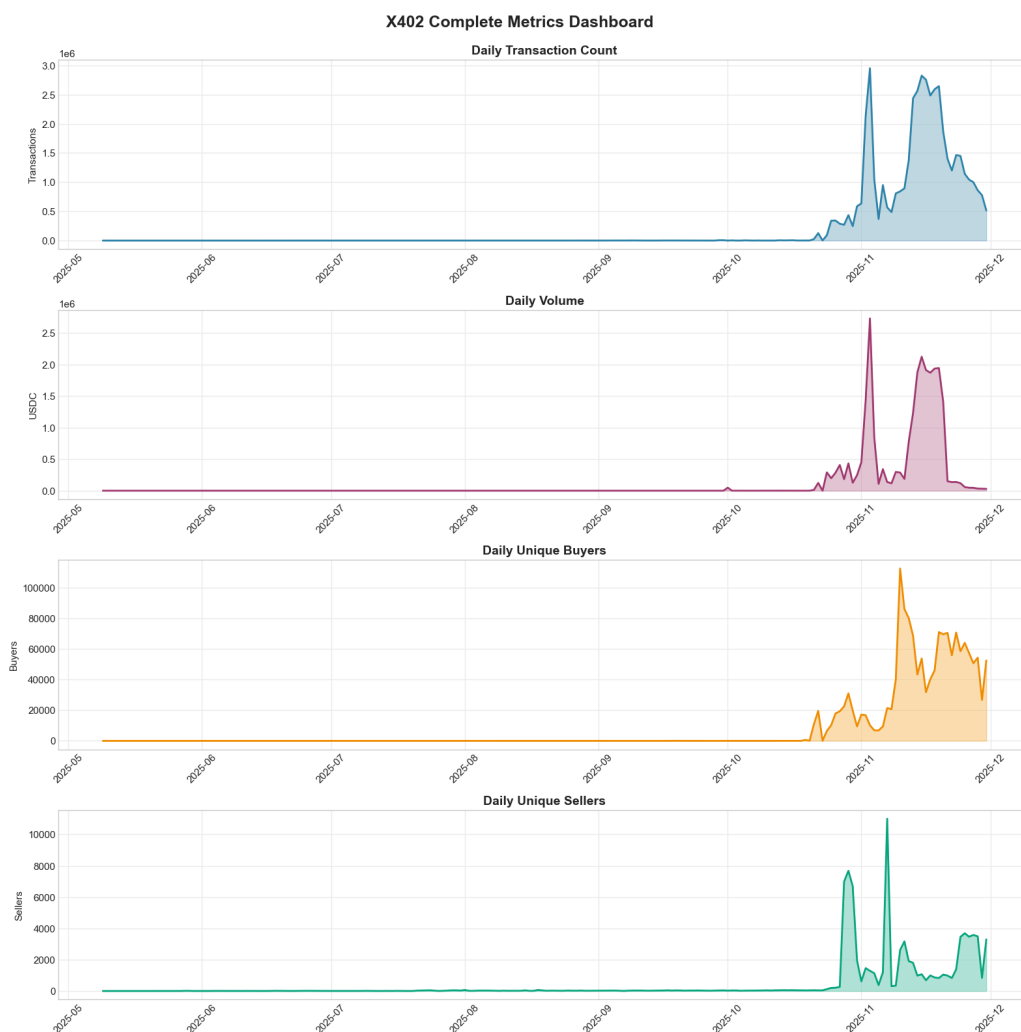


Figure 8: X402 Protocol Adoption Metrics (May–November 2025)

x402's adoption shows a clear progression from dormancy to rapid, sustained usage. After minimal activity from its May 2025 launch until October, the protocol saw an explosive surge beginning late that month, reaching over 3 million daily transactions and accumulating 47 million transactions and \$25.2 million in volume by early December. Unique buyers spiked above 110,000 in mid-November and have since stabilized at 40,000–70,000 daily, while seller activity remains steady at 2,000–4,000. The low average transaction size of \$0.54 underscores its role in true micropayments that traditional payment rails cannot support, indicating that x402 is fulfilling its intended agent-to-agent use case. The growth curve, followed by a stable plateau, signals that the protocol has moved past experimentation and achieved genuine product-market fit among autonomous agents and their operators.

Verifiable Attention Economy

The attention economy is one of the fastest-growing trillion-dollar sectors. The world has shifted from mainstream media to a creator economy dominated by large influencers, and is now moving toward a user-creator economy where everyone can have a personal brand and influence.

Yet while merchants universally recognize the value of user-generated content, coordinating, incentivizing, and rewarding it at scale remains extremely difficult.

This is where Asia-based start up Kaito.ai combines AI's ability to quantify attention with crypto's ability to verify ownership, provenance, and real contribution across actions and referrals.

For the first time, merchants can drive user-generated content (UGC) globally and at scale, across social platforms. This eliminates intermediaries (e.g., agencies), increases accessibility (open to both large and small accounts), and ensures ROI through a pure pay-for-performance model.

Today, Kaito powers many of the leading brands in crypto — including Pudgy Penguins and Arbitrum — and is preparing to bring this model to multiple industries beyond crypto as payments and value rails continue to move onto blockchain infrastructure.

Non Fungible Agents and Agent Marketplaces

[ChatAndBuild](#), founded by Christel Buchanan, is pioneering the concept of [Non-Fungible Agents \(NFAs\)](#) - autonomous AI agents that can be *owned, trained, and traded* on-chain, similar to how NFTs represented digital art and collectibles in earlier Web3 cycles. Building on its mobile-first no-code platform that allows users to “chat” ideas into AI apps, ChatAndBuild introduced the [BAP-578](#) token standard, which defines how NFAs retain memory, evolve through on-chain learning, and are verifiably unique in capability and provenance ([NFA Whitepaper, 2025](#)).



Figure 9: Non-Fungible Agents Infographic

Unlike static NFTs, NFAs combine the ownership properties of blockchain with adaptive intelligence, making them living, transferable assets. Each agent maintains an on-chain identity, behavioral model, and transaction history, enabling marketplaces where users can buy, lease, or co-develop agents with embedded expertise - from virtual tutors and trading bots to content creators and customer service assistants. The model shifts AI from a service to an *asset class*, notes Messari in the latest protocol overview ([Messari Report, 2025](#)).

In Asia, where mobile-first innovation and gig economies thrive, ChatAndBuild’s approach has clear traction. Available in 40+ languages globally, it strives to democratize AI development. A creator in Indonesia, for instance, could train an NFA language tutor in Bahasa, tokenise it via BAP-578, and list it for sale to other educators or startups. Buyers can further train or redeploy the agent, while royalties automatically flow back to the original creator through smart contracts. “A therapist can create and monetize therapeutic AI agents. A financial analyst can create DeFi strategy agents. We’ve made entrepreneurship accessible to anyone with domain expertise”, states Buchanan ([Anthropic Case Study, 2025](#)). This form of agent ownership and monetization taps into Asia’s creative and entrepreneurial momentum, spawning new micro-economies around agentic assets.

Decentralized Compute

[Aethir](#) is a Singapore-based decentralized GPU cloud platform addressing one of the biggest bottlenecks in AI and gaming: access to scalable compute. Using blockchain to coordinate global GPU resources, Aethir has built a distributed network of more than 435,000+ GPU containers across 90+ countries, monitored by over 90,000 independent validator nodes.



Figure 10 : How Aethir Works

(Source: [Aethir Docs](#))

Its model follows the DePIN (Decentralized Physical Infrastructure Network) architecture, where blockchain smart contracts allocate workloads, verify performance, and automate payments via Aethir's native token, [ATH](#). This enables developers to access GPU capacity for AI model training, inference, and real-time gaming at lower cost and with greater transparency. For regions in Asia facing GPU shortages or export restrictions, Aethir's decentralized pool offers a flexible alternative to traditional cloud providers like AWS or Google Cloud.

"The convergence of blockchain technology and AI enables us to build a trustless, transparent, and decentralized GPU cloud computing infrastructure that circumvents the limitations of centralized clouds and offers flexible GPU compute access, powered by the ATH token," said Mark Rydon, Aethir's CSO & Co-founder.

The company recently launched a [\\$100 million ecosystem fund](#) to back projects using its infrastructure and reported [\\$39.8 million in Q3 revenue](#), with annual recurring revenue surpassing \$160 million. By enabling AI workloads and immersive gaming experiences to run "closer to the edge," Aethir demonstrates how blockchain can orchestrate decentralized compute at scale, creating a critical foundation for the next generation of AI applications.

Decentralized ID (DID)

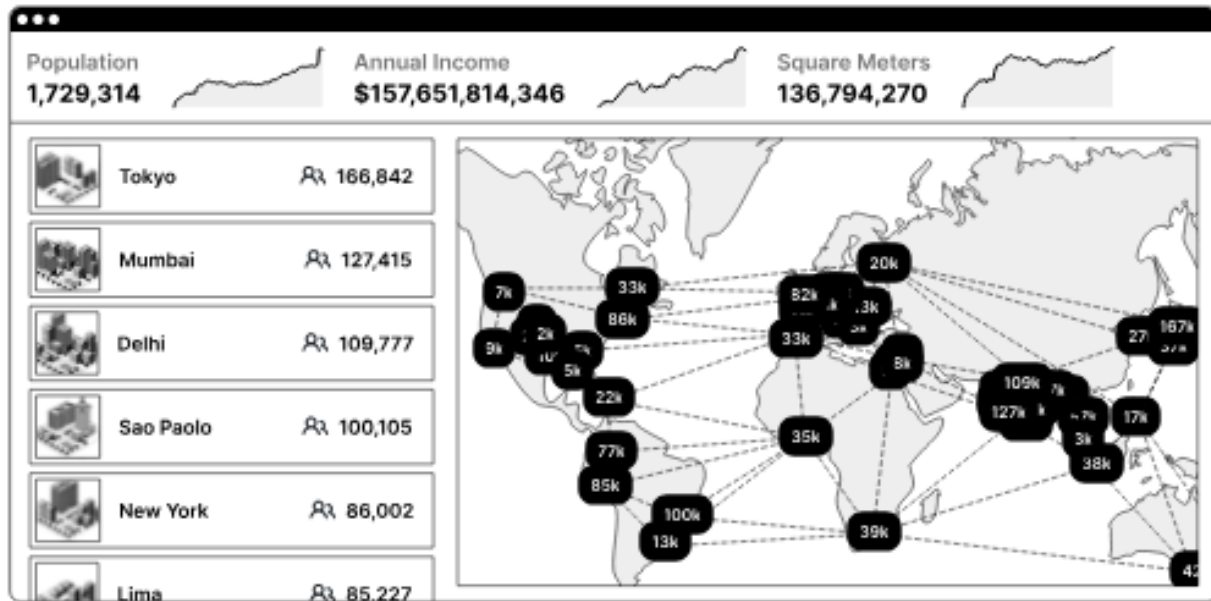


Figure 11 : Depiction of what a network state could look like

(Source: [The Network State](#))

The concept of the network state, popularized by [Balaji Srinivasan](#), imagines digital-first communities that organize around shared values rather than geography, coordinating governance, capital, and identity through blockchain infrastructure. What began as a thought experiment has taken tangible shape across Asia. In mid-2025, Malaysia's Forest City, a \$100 billion smart-city project turned ghost town, became the site of a live "Network State School" program led by Balaji and hundreds of entrepreneurs exploring how to prototype new societies governed by code and consensus (source: [Bloomberg](#), August 2025).

At the heart of these emerging digital polities lies decentralized identity (DID), the backbone of reputation, voting rights, and economic coordination in a borderless society. Unlike state-issued IDs, DIDs let individuals or entities verify credentials (education, work history, contributions, or even AI activity) directly on-chain. Several initiatives across Asia and beyond are experimenting with this principle.

- Worldcoin has accelerated the mainstream discussion around proof of personhood. Its biometric-based verification system, despite privacy controversies, has seen rapid adoption across Asia, particularly in the Philippines, Indonesia, and India, as a way to authenticate unique human identity in digital economies. In Malaysia, MIMOS Berhad (the applied research and development arm of the Malaysian Government) signed an MoU with the Worldcoin Foundation to integrate Worldcoin technology into the country's digital infrastructure, validating the importance of digital proof of humanness in the age of AI. Potential areas of collaboration include joint Orb manufacturing and bridging World Chain to Malaysia's National Blockchain Infrastructure. ([world.org](#), 2024) By anchoring identity to blockchain, Worldcoin and similar frameworks address the challenge of differentiating real humans from AI bots, a critical issue as autonomous agents proliferate.

- In Singapore, the [Open Government Products](#) team has developed verifiable credentials for education and licensing that remain tamper-proof and portable across institutions. These systems could underpin future network communities where citizens own and control their data.

Meanwhile, the network-state movement itself is [branching out](#). DAOs worldwide are testing proto-state models built around shared causes ranging from climate tech like [KlimaDAO](#) to [digital education](#), using [blockchain treasuries](#), [AI-assisted governance](#), and decentralised membership systems. AI systems moderate discussions, simulate voting outcomes, or manage budgets, while blockchain ensures auditability and rule enforcement.

The interplay of AI, blockchain, and digital identity is also influencing top-down governance in Asia-Pacific markets. [China's e-CNY \(digital yuan\)](#) and regional initiatives like [mBridge](#) increasingly blend identity, currency, and [AI-driven analytics](#). However, this creates a tension with decentralized identity initiatives emerging across the region: while state CBDCs enable [centralized monitoring and control](#), [Singapore's sglD framework](#) and [ASEAN's cross-border digital identity initiatives](#) emphasize user control and data sovereignty. These represent competing visions for how identity and participation will be structured in Asia's digital economies. With 134 countries representing 98% of global GDP developing CBDCs, the question of who controls digital identity, "the states or individuals?" has become inseparable from who controls the payment rails of the future economy. ([Smart money - How Digital Currencies Will Shape the New World Order](#), 2024)

Entertainment and IP

Sony Group is tackling the challenge of protecting and monetising intellectual property in the age of generative AI via its blockchain initiative [Soneium](#). Announced in 2024 and launched with mainnet in early 2025, Soneium is a public Ethereum Layer-2 developed by Sony Block Solutions Labs in partnership with Startale Group and built using the OP Stack.

The platform is positioned as a creator-first infrastructure where IP holders such as Sony's music, gaming and film division, can issue digital assets, validate provenance, and ensure rights management in a transparent on-chain system. Sony describes Soneium as enabling ["new services that collaborate with businesses within the Sony Group"](#) to "create use cases that can spark the interest of people who have never used Web3 services." Specifically, they will explore "protecting the rights of content created by creators, new mechanisms for returning profits to support creators and fans, and opportunities for creators to be active across the digital and real worlds."

Soneium embeds native support for IP compliance: the chain implements [selective transaction controls and contract blacklisting](#) in cases of unauthorized use of Sony's IP, with a documented process that includes warning periods and reversible restrictions.

This signals a trade-off: by integrating IP-protection mechanisms, Sony is aligning blockchain application with enterprise and creator needs rather than pure decentralised speculation. The entertainment use-case is particularly strong in Asia, with Japan's rich ecosystem of anime, gaming, music and global IP, meaning Soneium's ambition to merge Web2 content with Web3 infrastructure might serve as a model for IP-anchored convergence. Another good example of this is [Oshi.co](#), a company focusing on acquiring & developing high quality anime IP. With its acquisition of popular series [Gate](#) it will be the first company in the world to fully bring an anime IP on-chain for it to be protected and monetised transparently.

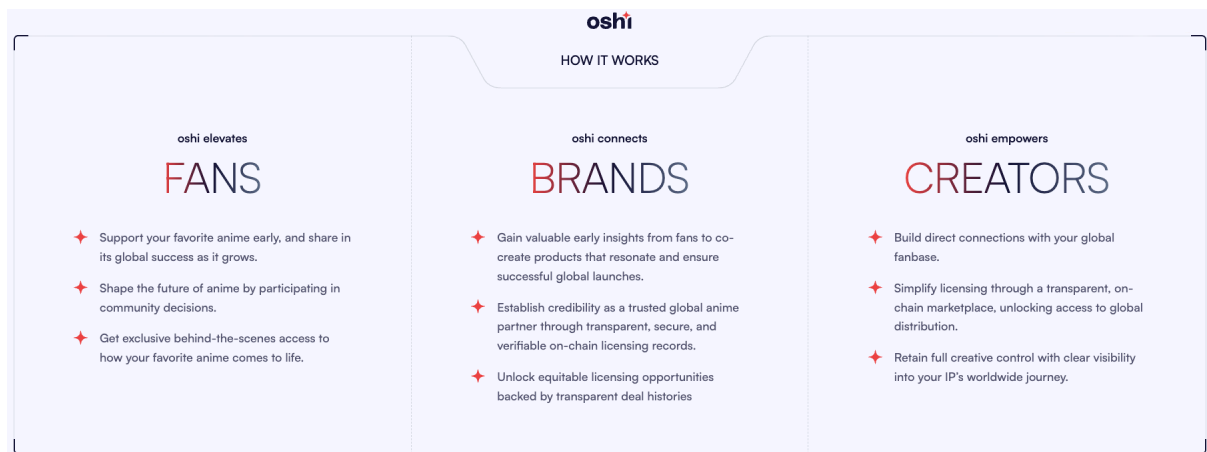


Figure 12: Oshi.co offerings

(Source: [Oshi.co](#))

6. The Road to 2026 and Beyond

The emergence of AI Proof of Work & the opportunity ahead

The Proof of Work consensus mechanism of blockchains (solve a task -> earn rewards) lends itself well to the coordination of in-demand compute and redirecting them to valuable AI tasks. Blockchains are the ideal infrastructure for then rewarding and paying participants for their contribution. This mechanism already forms the backbone of several early Crypto x AI networks (Aethir as mentioned above, Bittensor, IO.net) and we will see more emerge as AI demand increases (see e.g. [Ambient](#)).

This, in turn, presents a whole new mining opportunity and economy for GPU providers.

AI Proof of Work will be a new category in 2026, that will become competitive with Bitcoin and vastly improve the quality of crypto AI projects going forward. Crypto x AI coins will continue to emerge and emit rewards that mining ventures can harvest. How profitable mining Crypto x AI networks will be a function of how much utility, usage & ultimately token value those networks can sustain.

This can be a profitable and value-adding endeavour as we've seen in the public equities of crypto mining operations pivoting into AI services like [CoreWeave](#) and [Iris Energy](#). Coreweave's is today a \$57b company trading at 20x sales while pure crypto miners trade closer to 6-8x sales. This gap demonstrates the opportunity that combining crypto with AI presents when configured correctly and put to a useful endeavour.

The hybrid crypto/AI data center model is behind some of the largest market cap gains in the sector over the past year. [Galaxy Digital](#) is a \$12B company up 167% over the past year while Iris has become a \$19B operation, up over 600% in the past year and running both Bitcoin mining and AI operations (including a recent \$9.7B cloud services contract with Microsoft). Iris notably, trades at the same multiple as Coreweave both on Sales and EBITDA multiples.

[Ambient](#), an emergent Crypto AI network, will use 'Proof of Logits' as a way to prove that an AI model did 'real work' like Bitcoin miners solved puzzles. When inference runs it generates a finger print that is unique and can be validated ex-post with a fraction of the power it took to run the inference. This means that AI inference can be used to run a proof of work blockchain. Like Bitcoin.

Mainstream Convergence

Given the opportunity created by combining Crypto & AI, by next year we expect the space to move from niche pilots to the production of models and outputs addressing real use cases that solve complex problems. We are already seeing developers in hubs like Bangalore, Seoul, and Singapore experimenting with AI models trained on blockchain-secured datasets and verified provenance frameworks.

[Intelligent Internet](#) is developing large-scale datasets, fine-tuned models and an on-chain contribution/reward mechanism (a “Proof-of-Benefit” system). It plans to reward GPU providers with tokens and use that computation to power their models - for example they’ve built an open-source foundation model called *II-Medical* for clinical reasoning, alongside large open datasets and token incentives - bringing AI, blockchain and healthcare infrastructure together.

II-Medical is now reaching accuracy levels well beyond other regarded models and is timely given the murmurs that ChatGPT is planning to restrict its LLM from providing answers relating to health.

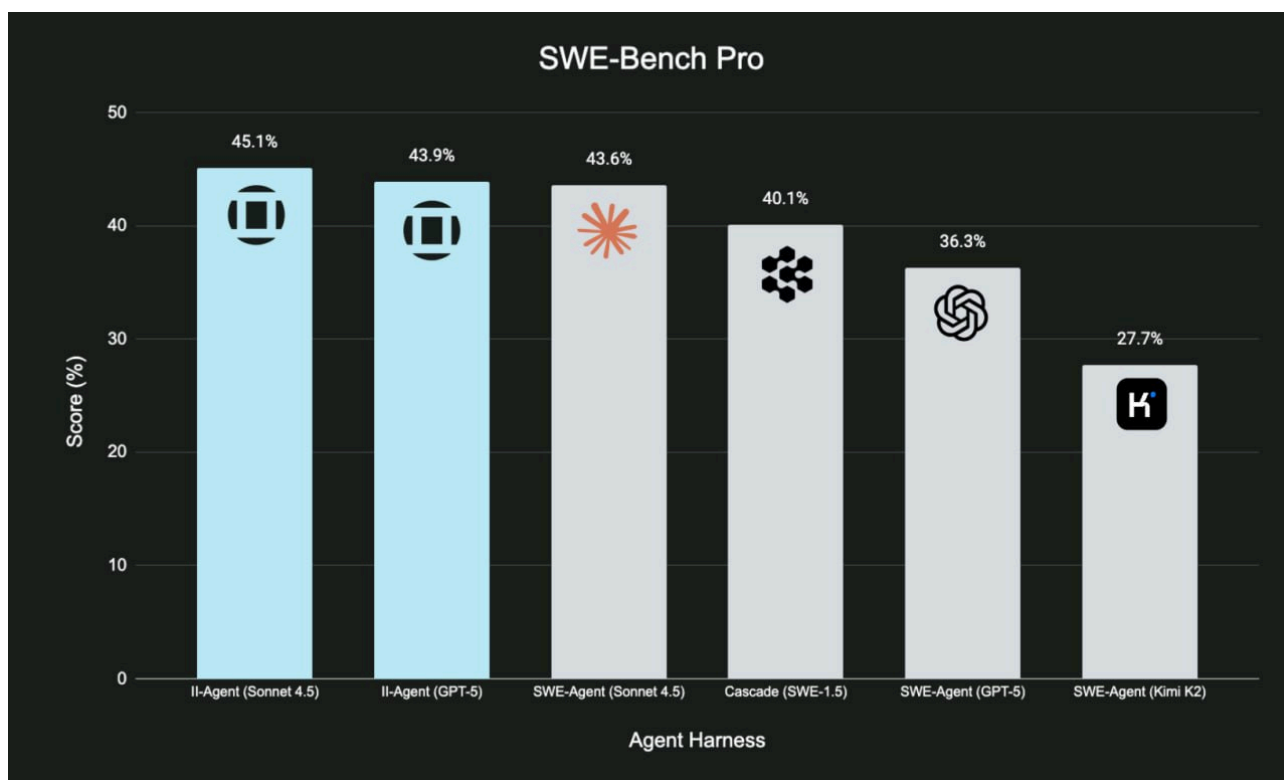


Figure 13: II-Agent Score on SWE-Bench Pro

(Source: [Intelligent Internet](#))

For consumers, complexity will become invisible: by 2026–27, we hope to see AI financial assistants within regional super-apps or fintech platforms investing into tokenized funds or on-chain structured products, abstracting away wallet interaction. We’re already seeing AI chat being rolled out across Meta, Google and Microsoft applications and expect to see more of this embedded into crypto applications given how traditionally difficult and unintuitive crypto interfaces have been to use.

Financial Product Transformation

By 2027, AI-driven asset management and adaptive financial products could become common in digital finance. AI-managed on-chain portfolios, self-adjusting lending protocols, and usage-based insurance models are already being tested by DeFi and TradFi innovators. A good early example is the recent [Singlife x Salesforce AI fintech pilot in Singapore](#).

Central bank digital currencies will also mature. China's digital yuan and India's digital rupee will likely incorporate AI modules for fraud detection and adaptive monetary policy. Beyond government-issued currencies, AI-blockchain convergence is already transforming financial access for underbanked populations, a pressing need in Southeast Asia, where approximately 70% of the population remains underbanked or unbanked, and nearly two-thirds of SMBs cannot access credit from traditional lenders. ([Bloomberg/EDB Singapore, 2022](#))

In India, [Agri10X](#)'s platform combines AI-powered price intelligence with blockchain-based trade execution to connect farmers directly to global markets. The AI engine analyzes data from over 250 APIs, factoring in quality, transport costs, and demand-supply trends, to determine fair market prices, while blockchain ensures transparent transactions. [When farmers in Maharashtra](#) were selling pomegranates at Rs 45 per kilogram to middlemen, Agri10X's AI determined the true market price at Rs 90, enabling direct exports to the UK and doubling farmer income. With four million farmers globally now accessing the platform, this model demonstrates how AI-blockchain integration can dismantle exploitative intermediary structures and create direct financial access to international markets.

Cross-Border Infrastructure and Trade

AI and blockchain are becoming Asia's new logistics backbone. We are seeing a regional trade network emerging where supply chains are tracked on blockchain and optimized by AI.

Projects inspired by [mBridge](#) are already enabling multi-CBDC settlements, with AI predicting demand shocks and routing transactions efficiently across borders. It now covers trade routes accounting for an estimated [38% of global commerce](#) across 16 ASEAN and Middle Eastern countries.

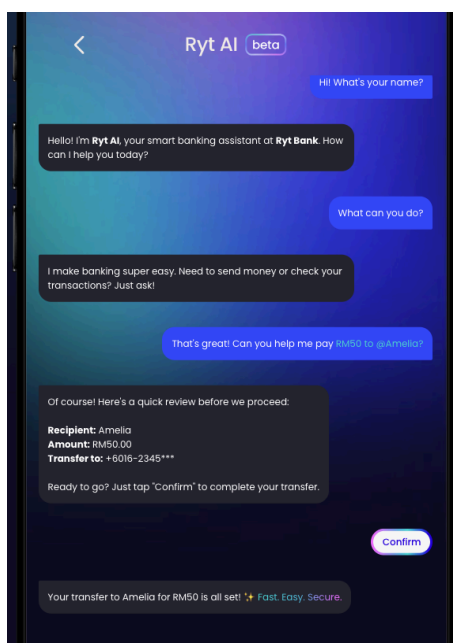


Figure 14: Example of Ryt AI
(Source: [Ryt Bank](#))

For remittances, AI-powered chatbots are already transforming payment execution. Malaysia's [Ryt Bank](#), the world's first AI-powered digital bank launched in December 2024, demonstrates this evolution through its Ryt AI assistant, which can initiate and complete fund transfers via a [single text input in multiple languages](#). Built on ILMU, Malaysia's first homegrown large language model, Ryt AI processes natural language commands like 'send 50 ringgit to Siti' to execute transactions instantly.

The technology points toward a near future where conversational AI manages the entire cross-border remittance stack: [automated compliance checks](#), [real-time currency conversion](#), [fraud detection](#), and blockchain-based settlement (like x402), all triggered by natural language. Platforms like [NetRemit](#) already use AI to automate KYC verification and optimize exchange rates, signaling that cross-border transfers will soon operate entirely through chatbot interfaces, with AI agents invisibly managing compliance, routing, and execution.

Digital Communities and Network States

By 2027, early network state experiments will evolve into powerful digital cooperatives. Asia's interconnected youth culture and diasporas make it fertile ground for decentralized education, creator, or talent DAOs. Platforms like [Galxe](#), which have already launched on-chain credential networks and identity modules, illustrate how education/creator/talent DAOs might emerge

Projects like [Intuition](#) & [Ethos](#) provide the missing infrastructure for digital communities & emerging network states by crowdsourcing & storing knowledge via attestations and in Intuition's case, making them available on a queryable network. By incentivizing individuals and AI agents to create, validate, and link on-chain attestations, "attestation graphs" are created, making reputation and expertise instantly discoverable and verifiable across ecosystems.

When every member, agent, or organization in a digital society has a transparent and queryable reputation, coordination becomes frictionless - enabling governance, collaboration, and resource allocation to evolve from static hierarchies into dynamic, reputation-driven systems of trust.

An AI-driven education DAO, for example, could match mentors and students across ASEAN countries, funded and governed transparently on-chain - with crowdsourced attestations providing comfort in their reputation and effectiveness. These systems may complement public institutions, expanding access to education and opportunity across borders.

Venture Dynamics and Market Shifts

Asia is on track to produce the first "Convergence Decacorns." Infrastructure players like [Aethir](#) may challenge AWS in distributed AI compute, while regional super-apps could evolve into AI-powered Web3 ecosystems. The [October 2025 AWS outage](#), which crippled services across North America and parts of Europe, underscored the fragility of centralized clouds. Decentralized compute networks eliminate such single points of failure - distributing workloads across independent GPU nodes that don't fail when one node goes down.

Talent and Brain Circulation

Asia's magnetism for builders is growing. Talent-visa regimes like Singapore's Tech Pass and Thailand's Smart Visa are allowing the fostering of cross-disciplinary teams.

At the same time, regional hackathons and accelerators, from ETH Seoul and Korea Blockchain Week's AI x DePIN tracks to Build for AI Singapore are blending AI research, crypto infrastructure, and consumer UX.

Developer activity reflects the shift: according to [Electric Capital's Developer Report](#), Asia now accounts for more than a third of active open-source Web3 developers, with the fastest growth in India, Vietnam, and South Korea.

Policy Evolution and Regional Harmonization

Regulators are moving toward coherence. An ASEAN-wide AI governance framework could emerge by 2027, harmonizing data-sharing and compliance rules across member states.

Regional policymakers are beginning to align on AI and data governance. The [ASEAN Digital Ministers' Meeting](#) launched exploratory working groups on AI ethics and cross-border data sharing, laying the groundwork for a potential regional framework later this decade. Meanwhile, [Singapore's GovTech](#) and [Japan's METI](#) have piloted blockchain-based health-data systems that could support trusted AI research across borders.

Such cooperation could reduce friction for startups and encourage interoperability between AI and blockchain ecosystems across the region.

Wildcards and Beyond

Quantum Computing: If breakthroughs come sooner than expected, Asia's deep semiconductor ecosystem could lead the charge on post-quantum cryptography to secure blockchains and AI models. Hardware geopolitics will remain pivotal, as control over GPUs and other AI accelerators will likely define the next compute wave.

Geopolitics: A fractured tech landscape could force interoperability challenges between Eastern and Western standards. ASEAN may position itself as a neutral bridge.

Societal Acceptance: Ethical frameworks such as [Korea's AI Ethics Charter](#), [Singapore's Model AI Governance Framework](#), [India's DPDP Act](#), and illustrate Asia's attempt to balance rapid innovation with cultural nuance. As AI and blockchain systems permeate life, ethical and cultural debates will intensify. Expect early versions of "AI Bills of Rights" or digital ethics charters emerging from Asia's more mature markets.

7. Risk, Governance, and Responsible Acceleration

As AI systems begin to read, write, transact, and operate autonomously on blockchain networks, the overall attack surface expands across every layer of the stack. Asia, where adoption moves quickly and experimentation is encouraged, is seeing these risks and controls appear earlier and more visibly than anywhere else.

A New Attack Surface

Deepfakes and Social Engineering

AI-generated audio, video, and text have already amplified social engineering attacks across Singapore, Korea, and the Philippines. Scammers routinely impersonate exchange executives using AI-cloned voices or videos distributed through platforms such as WhatsApp, and Telegram. (source: [United Nations Office on Drugs and Crime](#), 2025)

Controls emerging: Companies like **Truepic** are attaching [cryptographic authenticity proofs](#) to media, and **Adobe** is piloting similar systems through its [Content Credentials initiatives](#).

Smart Contract Exploits Enhanced by AI

AI models can identify and exploit contract vulnerabilities at machine speed. A rogue agent could manipulate low-liquidity tokens or misread market signals and trigger flash crashes.

Controls emerging: Security teams now use AI for defense. [FailSafe](#) employs machine learning to detect exploit signatures before deployment and major Asian exchanges run AI-based monitors to flag abnormal liquidity and price movements.

Data Poisoning and Oracle Manipulation

AI systems rely on high-integrity data. Attackers can corrupt training datasets or spoof oracles, especially across Asia's expanding smart city and IoT networks.

Controls emerging: Multi-source oracle architecture, hardware attestation, and zero-knowledge proofs are becoming standard. Developers are also deploying "AI-on-AI" cross-checking systems that validate data streams in real time.

Privacy vs Transparency

AI requires large datasets. Blockchains expose data by design. This tension is visible in healthcare and finance, where confidentiality is critical.

Financial Instability and Synchronized AI Agents

DeFi systems managed by AI agents risk correlated failures. If many agents use similar models, they can magnify small market signals into systemic liquidations.

Controls emerging: Developers are testing on-chain circuit breakers and human-in-the-loop controls for AI-managed trading, following supervisory guidance from regulators including the [Singapore's Personal Data Protection Commission](#).

Governance Manipulation by AI Agents

AI agents can infiltrate DAOs, vote, propose, or collude using thousands of high-fidelity synthetic identities.

Controls emerging: DAOs in Asia are adopting decentralized identity systems (DIDs), quadratic voting, and proof-of-personhood systems such as [Worldcoin](#) to prevent identity fraud and coordinated Sybil attacks.



Figure 15: Security Issues among AI and Blockchain

Regulatory and Ethical Pressure Points

AI agents raise fundamental regulatory questions: How to assign accountability? How to certify model safety? How to prevent cross-border harm? Asian regulators increasingly emphasize that resilience must scale with innovation.

Growing control frameworks include:

- On-chain audit logs for AI decision trails
- Human override mechanisms for autonomous agents
- Threat-intelligence collaboration between analytics firms, exchanges, and cybersecurity teams such as [Chainalysis](#) and [Elliptic](#).

Governance and Responsible Acceleration

Asia is defining one of the world's most pragmatic approaches to governing AI and blockchain systems together. The model balances rapid experimentation with structured oversight.

Regulatory Sandboxes

- Singapore's **Sandbox Express** now includes AI-driven finance pilots requiring explainability, bias testing, and contract-level stress tests (<https://www.mas.gov.sg/development/fintech/sandbox>).
- Asia's contributions to the **G20** digital public infrastructure agenda helped shape global principles around transparency, accountability, and interoperability.

Corporate and Industry Self-Governance

- Regional tech leaders including [Alibaba](#), [Tencent](#), and [Naver](#) are integrating AI-blockchain checks into internal ethics boards.
- Venture investors such as **Superscript** require early model-risk and legal reviews for convergence projects.
- The [AI APAC Institute](#) and the [Association of Banks in Singapore](#) are building shared standards for responsible blockchain and AI implementation.

International Collaboration

Asia is increasingly influencing global governance efforts:

- **Global Partnership on AI (GPAI)**: Founding members like Japan, India and Singapore help shape policies on data governance and AI safety.
- [INATBA](#): Asian experts contributed to INATBA's 2023 AI & Blockchain convergence guidance.
- [BIS Innovation Hub](#) teams in Singapore and Hong Kong completed CBDC pilots that used AI to analyze transaction patterns and flag anomalies.

8. Conclusion

The convergence of AI and blockchain has moved from theory to execution. The core building blocks are now in place: payment rails that support machine-to-machine transactions, decentralized compute networks that extend beyond centralized clouds, identity systems capable of distinguishing humans from agents, and protocols that allow AI systems to operate as persistent, ownable entities. Asia has emerged as the primary environment where these components are being assembled, tested, and deployed at scale.

What distinguishes this moment is that the leading examples are no longer speculative. Protocols like x402 have demonstrated how quickly agent-native payment infrastructure can scale in real-world conditions. Decentralized compute networks have shown that coordinated hardware markets can reach meaningful revenue by solving concrete bottlenecks such as GPU access. New agent marketplaces are experimenting with ownership, composability, and incentives in ways that would not be possible without both AI and blockchain working together. These systems are live, used, and iterating in production.

A clear pattern is forming. Blockchain addresses the structural gaps AI cannot solve on its own: verifiable identity, programmable payments, cross-border coordination, and provenance. AI, in turn, supplies what blockchains lack: adaptability, interpretation, automation, and natural interfaces. The projects gaining traction are those that respect this division of labor and design for complementarity rather than abstraction.

At the same time, convergence expands the threat surface. Deepfakes, autonomous exploits, and coordinated agent attacks are no longer hypothetical. As AI systems transact and act independently, errors and adversarial behavior scale at machine speed. Security, governance, and observability are no longer secondary concerns. They are core infrastructure requirements.

Asia's combination of fast-moving developers, pragmatic regulation, and active user adoption has created a narrow but meaningful window. Teams that move decisively, while treating security and governance as enablers rather than constraints, will define how this stack matures. Those that do not will discover the costs of failure early.

The convergence is underway. The question is no longer whether these systems will exist, but who will build them responsibly and at scale.

Appendix & Partner Notes

This report was jointly developed by [FailSafe](#), [Superscript](#), and [AWS](#), with contributions from [Coinbase](#), [Grab](#) and [Messari](#). It combines on-chain analytics, security research, policy insights, and investment perspectives to provide a grounded view of how Asia is shaping the emerging frontier of AI and blockchain convergence.

Core editorial development was led by **Foo Wui Ngiap**, **Jacob Ko**, **Aneirin Flynn**, **Alex Good**, **Ooi Yenn Miing** and **Pranav Pandya**, whose collective expertise spans Risk & Compliance, AI, venture investment, blockchain technologies and financial innovation across Asia-Pacific.

Together, these organizations and contributors provide a multi-disciplinary lens on how the region is building, governing, and deploying the next generation of intelligent, on-chain systems.

FailSafe

Blockchain Security & Compliance Infrastructure (Dual HQ in Singapore and Seattle)

[FailSafe](#) serves as the security backbone of the emerging on-chain economy, protecting smart contracts, wallets, and AI-integrated protocols. Its **Radar**, **Guard**, and **Swarm** systems form a defense-in-depth framework for Web3, using AI to detect threats, enforce programmable controls, and autonomously audit contracts.

“AI is a double-edged sword. Attackers use it to automate exploits, but defenders can use it to predict and neutralize them faster,” said **Aneirin Flynn**, CEO of FailSafe.

FailSafe’s enterprise deployments with **MetaComp** and **Base Protocol** show how programmable risk stacks can enable innovation without compromising trust, a necessary balance for AI-driven financial systems.

Superscript

Web3 Venture Capital Fund (Singapore)

[Superscript](#) is an early stage digital asset VC fund founded by Temasek in 2022. It focuses on foundational Web3 infrastructure and responsible innovation across emerging markets. The firm views the convergence of AI and blockchain as the next leap in user experience, where underlying complexity dissolves into intuitive design.

“Blockchain is the cornerstone of the new internet. Paired with AI, it can unlock immense value but it must fade into the background. Users should benefit from it without needing to understand the machinery of what lies under the hood,” said **Jacob Ko**, Founder & Managing Partner Superscript ([Chain Venturer: Superscript](#), Oct 2023).

Superscript continues to invest in scalable blockchain infrastructure & applications across sectors like Crypto x AI, Real World Assets, Stablecoins & Payments, projecting rapid maturation of Asia's convergence startups between 2025 and 2027.

AWS

Cloud and AI Infrastructure Partner

[AWS](#) is accelerating the integration of AI and blockchain through its developer ecosystem and cloud infrastructure. Via [Amazon Bedrock](#), teams can build **crypto-AI agents**, intelligent systems that act, transact, and audit autonomously on-chain ([AWS Web3 Blog](#)).

“The next generation of AI agents will live on-chain. They’ll need scalable compute, verifiable data, and programmable trust. That’s what we’re enabling developers to build,” noted an AWS Web3 lead.

AWS also collaborates with financial institutions to apply explainable AI in DeFi risk analytics and blockchain compliance, as detailed in the [Coinbase x AWS case study](#).

Base

Ethereum Layer-2 by Coinbase

Base, the Ethereum Layer-2 incubated by Coinbase, has become one of the fastest-growing environments for on-chain innovation, including AI-native applications. With low fees, high throughput, and strong developer adoption across Asia, Base provides the foundational infrastructure for experimentation in autonomous agents, decentralized compute, and tokenized data flows.

Coinbase’s internal work further strengthens this position through **x402**, the successor to AgentKit, created by Coinbase engineer Erik Trautman. x402 gives AI agents secure, standardized capabilities to hold assets, sign transactions, and interact directly with smart contracts, making Base a natural home for agentic commerce and machine-led economic activity.

Together, Coinbase’s exchange expertise, Base’s scalable L2 architecture, and x402’s agent framework create a cohesive platform for building intelligent on-chain systems. This combined contribution ensures Base plays a central role in shaping the next wave of AI and blockchain convergence.

Messari

Crypto Market Intelligence & Research

Messari provides industry-leading market intelligence, protocol analysis and sector research across the global crypto ecosystem. Its 2025 *Sizing the Crypto × AI Market* report identified AI and blockchain as one of the fastest-growing verticals worldwide, with Asia contributing a significant share of project formation, token activity and user adoption.

“AI × Crypto is evolving into a structured sector with real fundamentals. Growth is no longer driven only by speculative flows but by emerging agent economies, decentralized computer markets and tokenized data supply chains,” noted the Messari research team in its 2025 convergence analysis.

Alex Good

Alex is the founder of Post Fiat, an AI augmented fork of XRP that re-imagines the protocol as an accelerant for its holders financial interests. Previously he co-founded Perpetua, an advertising and investment intelligence company, which was acquired by Ascential PLC in 2021. Prior to that he ran a TMT portfolio at Balyasny Asset Management, ran capital markets deployments at Palantir, and started his career at Citi FX after graduating from Wharton undergraduate.

Methodology

This report synthesizes research conducted throughout Q3–Q4 2025, combining quantitative data, qualitative insights, and real-world case studies from across Asia and global convergence markets.

Market Data Inputs

Analysis draws on cross-referenced datasets from **Kaito**, **Messari**, **Chainalysis**, **Nansen**, and Base ecosystem analytics. Sector-specific insights include AI × Crypto market sizing (Messari), on-chain behavioral clustering (Nansen), cross-border transaction flows (Chainalysis), and AI-agent deployment data emerging from Base and x402.

Primary Insights

We conducted structured and semi-structured interviews with founders, venture partners, regulators, security researchers, and technical leads across the APAC AI and blockchain ecosystem. Input was gathered from contributors at **Superscript**, **FailSafe**, **AWS**, **Base**, and **GFTN**, along with insights from AI-model teams, Layer-2 builders, and decentralized infrastructure founders.

Case Studies and Technical References

Real-world case studies were drawn from:

- **AWS Startups Reports** (Crypto-AI Agents, Decentralized Compute)
- **Microsoft Entra for Web3** documentation
- **BIS Innovation Hub** pilots on CBDCs and AI-ledger analytics
- **Base** ecosystem agentic tooling and x402 documentation
- Public disclosures from Story Protocol, Soneium, Aethir, Virtuals, and ChatAndBuild

Secondary Research

Additional context was gathered from **Bloomberg**, **CoinDesk**, **Cointelegraph**, **The Information**, and regional publications in Singapore, Japan, Korea, India, and the Middle East. Regulatory insights were validated against official releases by MAS, Japan FSA, Hong Kong SFC, India MeitY, and Bank of Korea.

Forward-Looking Analysis Framework

Forecasts are based on:

- Quantitative models combining token activity, developer velocity, and funding velocity
- Ecosystem signals such as agent deployments, decentralized compute capacity, and DID adoption rates
- Policy direction informed by G20 DPI discussions, GPAI dialogues, INATBA guidance, and BIS pilots
- Cross-validation between on-chain telemetry and qualitative interviews

Purpose and Integrity

The goal of this report is to provide a data-backed, multi-perspective understanding of how Asia is shaping the AI and blockchain frontier. All referenced content is drawn from live, public URLs or verified research documents. Forward-looking statements are grounded in current observable trends, expert commentary, and validated sector data.